

Venture Capital and Entrepreneurial Management MGMT 264

Liquidation Preferences and Anti-Dilution Formulas Caselette # 4

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HiTech Capitalization Summary

Type of Stock	Common	Series A	Series B
Number of Share	3,000,000 (including 600,000 outstanding unvested options and 400,000 future option reserve)	Good Guys: 2,000,000 ; Wannabe: 1,000,000	World Venture: 2,000,000 Series A investors: 500,000
Share Price	N/A	\$3.00	\$2.00
Total Investment	N/A	\$3/share * 3,000,000 = \$9,000,000	\$2/share * 2,500,000 = \$5,000,000
BOD	2 seats	2 seats (1 for Good Guys and 1 for Wannabe)	1 seat
Liquidation Preference	None	1X Participating	4X Non-Participating Super Liquidation Pref.
Anti-Dilution	None	Broad-based Weighted Average	Full Ratchet
Others	None	50% of series A voting as a separate class are required for the sale or merger of Hi Tech	None

Q2: Valuation Post Series B Financing

Type of Stock	Number of Shares	Total Investment	Share Price
Common Stock	2,000,000 outstanding, 600,000 option grants, 400,000 option reserve	N/A	N/A
Series A Preferred	3,000,000 outstanding	\$9,000,000	\$3.00/share
Series B Preferred	2,500,000 outstanding	\$5,000,000	\$2.00/share

Question: Which numbers are relevant in calculating valuation upon Series B preferred financing?

Valuation = Total number of shares * Share price of Series B Preferred
 = (2,000,000 + 600,000 + 400,000 + 3,000,000 + 2,500,000) * \$2.00/share = **\$17,000,000**

Q3: Equity Ownership Percentage

Type of Stock	Number of Shares	Percentage of Equity
Common Stock	2,000,000 outstanding, 600,000 option grants, 400,000 option reserve	35%
Series A Preferred	3,000,000 outstanding	35%
Series B Preferred	2,500,000 outstanding	30%
Total Number of Shares	8,500,000	100%

Q4: Sale of HiTech for \$50 Million

■ Before Series B Preferred Financing

Series A Preferred: 1X Fully Participating Liquidation Preference

Get Liquidation Preference AND Participate with Common Stock Holders;

**NO NEED TO
MAKE A
CHOICE!!!!!!**

Get Liquidation Preference

Series A: 1X
1*\$9,000,000=\$9,000,000

Common Stock: Nothing

Firm value (after
liquidation
preference
realization) = \$50M-
\$9M=\$41M

*3M/(3M+2M)=3M/5M=60%

**2M/(3M+2M)=2M/5M=40%

Only shares that
are actually
outstanding count.
Unexercised
options or reserves
for future option
grants have no
relevance

Participate with common shareholders on a 1:1 converted basis

Series A: 60%**
60%*\$41M=\$24.6M

Common Stock: 40%**
40%*\$41M=\$16.4M

Q4 (Continued) – Participating Security with 1X Liq. Pref.

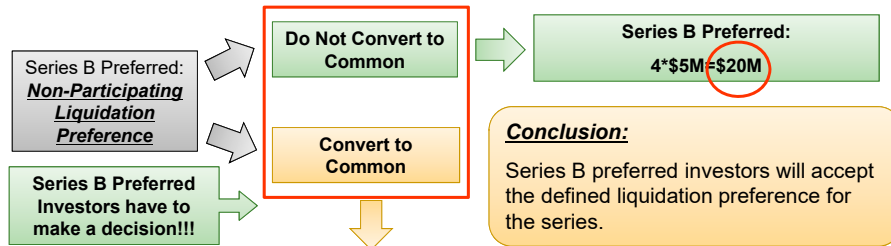
■ Economics between Series A investors and founders

Stock	Shares	% Own	Invest	Liq. Pref	Liq. Pref. Payout	Participate	Dist.	% of Dist.
Common	2M	40%	N/A	N/A	0	40%*\$41M = \$16.4M	\$16.4M	32.8%
Series A Preferred	3M	60%	\$9M	1X	\$9M	60%*\$41M = \$24.6M	\$33.6M	67.2%
Total	5M	100%	Firm Value:\$50M		\$9M	\$41M	\$50M	100%

Note: "Participating liquidation preference" will never face a situation where it is better to convert to common stock.

Q5: Sale of HiTech for \$50 Million (After Series B Financing)

■ Dynamics of Non-Participating Liquidation Preference



Stock	Shares	% Own	Invest	LP	Payout	Participate	Final	Final %
Common	2M	27%				$27\% \times \$41M = \$11.1M$	\$11.1M	22.2%
Series A Preferred	3M	40%	\$9M	1X	\$9M	$40\% \times \$41M = \$16.4M$	\$25.4M	50.8%
Series B Preferred	2.5M	33%	\$5M	4X	0	$33\% \times \$41M = \$13.5M$	\$13.5M	27%
Total	7.5M	100%	Firm Value: \$50M		\$9M	\$41M	\$50M	100%

Q5 (continued): Impact of Liq. Preferences

■ Economics between investors and founders

Stock	Shares	% Own	Invest	Liq. Pref.	Liq. Pref. Payout	Participation	\$ Dist.	% of Dist.
Common	2M	27%				$40\% \times \$21M = \$8.4M$	\$8.4M	16.8%
Series A Preferred	3M	40%	\$9M	1X	\$9M	$60\% \times \$21M = \$12.6M$	\$21.6M	43.2%
Series B Preferred	2.5M	33%	\$5M	4X	\$20M	0	\$20M	40%
Total	7.5M	100%	Firm Value: \$50M		\$29M	\$21M	\$50M	100%

Q7: Price Protection Anti-Dilution Formula

■ “Broad-based” Anti-Dilution Protection

Revised Conversion Price = Conversion Price * F

$$F = \frac{(\text{\# of existing shares outstanding} + \text{\# of existing options outstanding} + \frac{\text{investment}}{\text{old share price}})}{(\text{\# of existing shares outstanding} + \text{\# of existing options outstanding} + \frac{\text{investment}}{\text{new share price}})}$$

= \$3.00/share * $\frac{(2,000,000 + 3,000,000 + 600,000 + \frac{\$5,000,000}{\$3.00/\text{share}})}{(2,000,000 + 3,000,000 + 600,000 + \frac{\$5,000,000}{\$2.00/\text{share}})}$

= \$3.00/share * 0.89712 = \$2.69/share

of new shares that the new investor **would have received** for the amount of new money that is invested @ **old price/share**

of new shares that the new investor **will actually receive** for the amount of new money that is invested @ **new price/share**

$$\text{Conversion Ratio} = \frac{\text{Purchase Price}}{\text{Revised Conversion Price}} = \frac{\$3.00/\text{share}}{\$2.69/\text{share}} = 1.115$$

Q7: Anti-Dilution Formula

■ New Capitalization of the Company

Category of stock	Actual # of shares of stock	# shares as-if-converted to common
Common stock outstanding	2,000,000	2,000,000
Outstanding options (unvested)	600,000	600,000
Future option reserve	400,000	400,000
Series A Preferred	3,000,000	<u>3,345,725</u>
Series B Preferred	2,500,000	2,500,000

Note:

Converted Shares = Existing shares * Conversion Ratio = 3,000,000 * 1.115 = 3,345,725

Q8: Anti-Dilution Formula

- “Full Ratchet” Anti-Dilution Protection

Revised Conversion Price = New Price = \$1.00/share

$$\text{Conversion Ratio} = \frac{\text{Purchase Price}}{\text{Revised Conversion Price}} = \frac{\$2.00/\text{share}}{\$1.00/\text{share}} = 2$$

$$\begin{aligned}\text{Converted Shares} &= \text{Existing Shares} * \text{Conversion Ratio} \\ &= 2,500,000 * 2 = 5,000,000\end{aligned}$$

Note: In a full ratchet, it is as if the Series B Preferred made their \$5 million investment based on \$1.00 per share price – instead of getting 2.5 million common shares upon conversion, the Series B Preferred would be convertible into 5 million shares of common stock.

Your Questions?

